Subhrajit Mishra

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Education

May 2024 **Bachelor of Technology**, Department of Electrical Engineering, IIT Kanpur, India — *CPI: 9.2/10*

May 2020 **Grade XII (CBSE)**, Yogiraj Public School, Kota, India — *Percentage: 95.0%*

May 2018 **Grade X (CBSE)**, DAV Public School, Chandrasekharpur, Bhubaneswar, India — *Percentage: 97.2%*

Achievements

Jan 2025 **AI expert and Software Pro** for exceptional deep learning and software development skills — *Samsung*

Apr 2024 **Academic Excellence Award** for exceptional performance in the year 2021, 2023, & 2024 — *IIT Kanpur*

Mar 2022 **InterIIT Techmeet 10, Silver Medal** Aerial Vehicle guided Ground vehicle navigation — *IIT Kanpur*

Mar 2020 **KVPY Scholarship**, Ministry of Science & Technology, Govt. of India, among 268K applicants — *India*

May 2018 **NTSE scholarship** , Ministry of HRD, Govt. of India, among 1.4M applicants — *India*

Sept 2017 **INJSO and IAOSP**, National level qualifying stage for IJSO and IOAA — *India*

Research Interests

Machine Learning · Deep Learning · Computer Vision · Image and Video Signal Processing · Robotics · Perception

Patents

Methods and system for AI-based pixel-block prediction for video codec — Filed

Developed a unified, low-complex neural network architecture for intra prediction, which leverages neighboring reference lines to predict coding units (CUs) of **variable sizes** and is optimized for **efficient hardware** implementation.

NN-Based Reference Sample Enhancement for Intra Prediction — Filed

Developed an **enhancement network** to improve the input to intra prediction by enhancing the already decoded reference pixels, thus improving **compression efficiency**.

Adaptive Visual Artifact Removal in Video Codec using AI Inloop Filters — Filed

Developed a **content-based partitioning** of separate models for Neural Network based loop filtering which provides better subjective and objective **video quality**

Inverted-L-Scan Method for AI-based Intra-frame prediction in video codec — Filed

Developed a **pre- and post-processing** method utilizing an inverted L-scan order to prepare neighboring pixels for a **unified** AI-based intra prediction model within a post-VVC codec, resulting in enhanced coding efficiency.

Industry Experience

Samsung Research | Advanced Engineer — Bengaluru, India

Next Generation Video Compression Technology (Full-Time) — *July 2024 - present*

- **Neural Network Video Codec Development for Upcoming standard H.267:** Development and integration of NN-based algorithms for In-loop filters and Intra Prediction over existing standard, H.266 - Versatile Video Codec
- **Advanced Professional Video:** Developing perceptually lossless codec for **ultrasound probe** and pro video editor
- **MPEG-5: EVC and FFMPEG:** Open source codec development, core intrinsic optimization and on-device integration

EVERSANA: Asia-Pacific — Mumbai, India

Digital and Data solutions (Internship) — *May 2023 - July 2023*

- Conceptualization of **Willingness to Recommend** (WTR) and **Intent to pay** (ITP) for medical consumables
- Uncovering adequate **clusters** of physicians and hospitals as per behavioral implications and habitual practices
- Identifying **optimal interventions** for cohorts and creating a software of the solution to be used by MRs of client

Research Experience

Neural Network based Intra Prediction
Samsung Research

- Aim: Construct **unified** NN-based intra-frame prediction handling **varying input** and output Coding unit sizes
- Designed an **adaptable** and **configurable** CNN based network with an intelligent pre- and post-processing stages
- Integrated the NN-based intra-prediction mode **alongside** classical predictors in post-VVC, enabling C++ inference
- Replaced seven separate post-VVC model variants and achieved **State of Art** BD-BR gains at 1KFlops/pix complexity

AI-based In-Loop Filters in post VVC
Samsung Research

- Design low-complex **hardware-integrable** neural network for suppressing compression and banding **artefacts**
- Combined artefact-aware **handcrafted filters** with custom designed **attention** framework and conv core block
- Delivered state-of-the-art Bjøntegaard Delta Bitrate (BD-BR) gains at **ultra-low complexity** less than 500FLOPs/pix
- Integrated the model within the VVC Test Model, with **dedicated signalling** to operate in parallel to Deblocking filter

Open Advanced Professional Video (APV) Codec
Samsung Research

- Contributed core algorithms for APV, an open-source, **perceptually lossless** professional-grade video codec
- Optimized codec encoding and decoding speed by using **platform-specific** intrinsic tools (**NEON** and **AVX**)
- Integrated APV codec on Samsung devices and Premiere Pro for **National Association of Broadcasters**(NAB) demo

Multimodal Contrastive Learning - Undergraduate Project
IIT Kanpur | Code

Supervisor: Dr. Tushar Sandhan – *Perception and Intelligence Lab, IIT Kanpur* — *Jan - Mar 2024*

- Aim: Diagnosis of **Myocardial infarction** and **Coronary artery disease** using tabular and imaging patient data
- Designed a generalizable **SimCLR** based **contrastive learning multimodal** framework with **InfoNCE** loss
- Generalized the framework to **Medical equipment classification** for FDA/CDSCO using images and description
- Created an unimodal model for equipment classification with tabular data using **BERT** to compare performance

Academic Projects

Unmanned Aerial Vehicle (UAV) guided Unmanned Ground Vehicle (UGV) navigation
IIT Kanpur | Code

InterIIT TechMeet 10.0, DRDO challenge, *Problem statement and solution documentation* — *Feb - Mar 2024*

- **Autonomous traversal:** Incorporated a **PID velocity Controller** for modulating speed and a **Pure Pursuit** Controller for steering in the UGV for traversal on mountainous terrains, using odometry obtained by a hovering UAV
- **Real-time motion planning:** UAV camera image **segmentation** to identify road, the center and orientation of UGV
- **Autonomous takeoff of UAV and mean path detection for UGV:** Ensured minimal derouting from mean path

Atari Games with Reinforcement Learning
IIT Kanpur | Code

Supervisor: Dr. Subrahmanya Swamy Peruru – *Department of Electrical Engineering, IIT Kanpur* — *Mar - Apr 2024*

- Created a **DQN** and a **DDQN** model of five layers with the **cart pole environment** for playing **Atari games**
- **Results:** Model was run for two games: **Breakout & Pong**; it outperformed computer, rewards converging at max
- Added **LSTM** for **DRQN** model, achieving SOTA results for **partial observability** in **flickering** game environment

Antenna Designing for Communication Systems
IIT Kanpur | Code

Supervisor: Dr. Raghvendra Kumar Chaudhary – *Radio Frequency, mmWave Lab, IIT Kanpur* — *Mar - May 2024*

- Produced mmWave **horn antennas** for 75-110GHz, with **frequency selective surfaces** to control cross polarization
- Simulated designs of various types of antennas, finding optimal structures for best gains, **directivity** and **S plots**
- **Impact:** Designed **X-Band Mono-pulse** Conical Feed Antenna for **Low Earth Orbit** (LEO) Satellite Ground Station

Machine Learning Course Project
IIT Kanpur | Code

Supervisor: Dr. Purushottam Kar – *Department of Computer Science Engineering, IIT Kanpur* — *Jan - Apr 2023*

- Created **linear models** to crack simple & advanced **XORRO PUF** with >99% accuracy at ultra-low complexity
- Developed a Decision tree based model to play a hybrid of **Hangman** and **Wordle** with average query rate of 3.99
- Created an **electrochemical air quality sensor caliberator** using **Elastic Net** and **Extra tree regressor**

Skills

Programming & Web:	Python, C++, C, MATLAB, HTML/CSS, Solidity, Object-oriented programming, MySQL
Tech & Framework:	Linux, Github, ROS, Gazebo, Scikit-learn, Open CV, Excel-VBA
Softwares & Utilities:	Git, Numpy, Pandas, Arduino, bash, LaTeX, Microcap, Visual Studio

Test Scores

Aug 2025 **TOEFL iBT** Score: 112/120 *ETS*

Sept 2025 **GRE General Test** Score: 325/340 (Breakup - Quant: 170/170; Verbal: 155/170) *ETS*

Relevant Coursework

Artificial Intelligence	Advanced Machine Learning*, Deep Learning*, Reinforcement Learning, Conv Neural Nets* Computer Vision*, Sequence Models*, Probability and Statistics, Stochastic Process
[1] Computer Science	Introduction to programming, Data structures and Algorithms, Computer Organization*, Operating System*, Computer Networking*, Database management system*
Electrical Engineering	Signal Processing, Image Processing, Digital Electronics, Microelectronics, Principles of Communications, Electromagnetic theory, Electrical Engineering Labs - I & II
Mathematics	Real Analysis, Complex analysis, Linear Algebra, Ordinary and Partial Differential Equations

* - Online Courses

Positions of Responsibility

Team Head — Aerial Robotics, IIT Kanpur May '21 - Aug '23

- Learnt **robot programming** and Development of **drones** with simulation environments and hardware testing
- **Represented IIT Kanpur** at various national and international competitions including **InterIIT Techmeets**
- Led a two-tier team of **30 members** enabling **participation & sponsorships** for several **Aerial Robotics** events

Secretary — Entrepreneurship Cell, IIT Kanpur Apr '21 - May '22

- Spearheaded a three tier team of **100 students** for the successful conduction of **Entrepreneurship Summit**, 2022
- **Finalized sponsorship deals** and signed MoUs with various companies and continued as their first point of contact

Teaching Experience

Team Head — Aerial Robotics, IIT Kanpur Aug '22 - Nov '22

- Recruited **25 Junior members** for my team with induction to the required tech stack and simulation environments
- Mentored 7 junior members in a **mini-project** on **swarm robotics** guiding them in drone flight control & automation

Project Guide — Stamatics Society, IIT Kanpur May '23 - Aug '23

- Guided around **98 Undergrads** to master **Image processing algorithms** and utilizing them for computer vision
- Spearheaded **diverse projects** curated to depict the effect of classical image processing in Deep learning

Academic Mentor — Counselling Service, IIT Kanpur Aug '21 - Mar '22

- Conducted various **academic classes** for the course PHY102, teaching various topics to around **600 students**
- Personally guided **20+ freshman students** throughout their **first year** at all **academic fronts**

Volunteering and Extra-curriculars

- Spearheaded various **state level science exhibitions** and **Community Service Camps** as a commanding member of the civil defence service unit, **National Service Scheme** (NSS)
- Have been a regular **blood donor** with active involvement in organizing blood donation camps, since Oct' 21
- Active participant in **Competitive Programming** solving **700+** problems with a maximum rating of **1703**
- Member of **Debating Society and Bicycling Society, IIT Kanpur**, winning various **inter-collegiate** competitions
- Victoriously led the **Music, Football and Quizzing teams** in several **inter-college** and **inter-school** events